

Scott Okuma

Spacecraft Software Architect | Engineering Leader

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Summary

Spacecraft software architect and engineering leader specializing in ARM-based flight computers, embedded Linux, and autonomous recovery. Umbra's first flight-software engineer, with experience building the software organization, leading the transition from bare-metal to Linux, and delivering flight software across three spacecraft blocks deployed to orbit. Architected TRL 9 remote-update and redundant-computing capabilities for spacecraft operating without physical access. Electrical-engineering background includes FPGA-based S-band radio development, software-defined radio, RF systems, and spacecraft communications analysis.

Professional Experience

Umbra Space

Spacecraft Systems Architect | Goleta, CA | March 2025 to Present

Director of Spacecraft Software | Santa Barbara, CA | August 2021 to March 2025

Flight Software Lead | Santa Barbara, CA | June 2019 to September 2021

Selected accomplishments across Umbra roles

- Built Umbra's flight-software department from the ground up, hiring and managing 12 direct reports and establishing software architecture, development processes, nightly CI/CD, test infrastructure, and delivery practices.
- Drove spacecraft platform architecture in collaboration with approximately 20 electrical and mechanical engineers, spanning flight computers, GNC, payload, radio, avionics, and ground-command interfaces.
- Led the transition from bare-metal flight software to embedded Linux, with responsibilities spanning firmware, FPGA, Linux kernel development, and GNC model integration. Hands-on Zynq-7000 and Zynq UltraScale+ work included bootloaders, BSPs, drivers, DDR, DMA, and processor/FPGA interfaces.
- Architected two TRL 9 spacecraft update solutions: golden-image recovery for a Zynq-7000 controller and delta/rolling bootloader, kernel, and application updates across four onboard computers, incorporating nonvolatile staging, checksum validation, redundant boot images, and automatic rollback.
- Architected TRL 9 dual-flight-computer redundancy with watchdog supervision, onboard-computer selection logic, and automatic failover.
- Designed fault detection, isolation, and recovery (FDIR) logic supporting selective sensor/actuator resets, GNC model restart, and a low-power sun-pointing safe mode.
- Made hardware/software architecture and component-selection decisions for radiation-tolerant spacecraft systems, including nonvolatile-memory selection with FRAM.
- Delivered flight software across three spacecraft blocks successfully deployed to orbit, with no mission losses attributable to flight software.
- Developed environmental test plans spanning execution, teardown, and pre-/post-test functional validation; supported spacecraft thermal-vacuum, shock, and vibration verification.
- Reviewed processor schematics and PCB layouts to verify pin assignments and required hardware/software interfaces.

Earlier Experience

Electrical/Firmware Engineer

Raytheon - Goleta, CA | September 2017 to June 2019

- Designed digital firmware for RF system controllers using VHDL/Verilog and Vivado; customized Linux kernels and developed embedded systems with PetaLinux and C/C++.
- Conducted hardware checkout, schematic analysis, and DDR3 implementation for electronic modules, using oscilloscopes, spectrum analyzers, and network analyzers.

Electrical Engineer

NAWCWD - Point Mugu, CA | January 2015 to September 2017

- Developed software and performed RF hardware repair, layout, radar calibration, and test planning for radar systems.
- Created a DRFM jammer using an FPGA, RF transceiver, and user interface; programmed in Python, C++, and MATLAB for signal processing and simulation.

Electrical/RF Engineer

Computer Sciences Corporation - Point Mugu, CA | January 2012 to January 2015

- Developed automated test procedures for GaN RF amplifiers using LabVIEW and GPIB; built custom test fixtures and collaborated with technicians to troubleshoot systems.
- Verified equipment thermal limits using environmental chambers and data logging; analyzed test results using Octave.

University Projects

CSUN CubeSat1 - Satellite Communications Team Member

Jet Propulsion Laboratory / CSU Northridge | January 2014 to December 2014

Developed radio communication test procedures for electrostatic-sensitive devices and bit-error-rate, link-budget, and data-volume analyses for mission planning.

NOAA Radio Weather Receiver Team Member

CSU Northridge | February 2014 to June 2014

Designed an RF amplifier for a superheterodyne receiver and developed a software-defined radio using MATLAB/Simulink.

Technical Skills

Platforms: ARM; Zynq-7000; Zynq UltraScale+; FPGA design; DDR; DMA; SPI; UART; Ethernet; VPX.

Software: C; C++; Python; Rust; VHDL/Verilog; MATLAB/Simulink; LabVIEW; bare-metal; embedded Linux; RTOS; PetaLinux; Yocto; Vivado/Vitis.

Communications: S-band radios; software-defined radio; RF system design and test; link budgets; bit-error-rate and data-volume analysis.

Reliability: FDIR; remote updates and rollback; redundant flight computers; watchdog recovery; nonvolatile-memory selection, including FRAM; environmental verification.

Delivery: Git/GitHub/GitLab; CI/CD; test infrastructure; avionics integration; technical project management; hiring, leadership, and mentoring.

Education and Training

Bachelor of Science in Engineering | California State University Northridge | 2012-2014

Xilinx Embedded Design with PetaLinux Tools | November 2018